How Certain Is the Chatbot Arena Leaderboard?

Multiplicity, Rank Spreads and Category Specialties

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Abstract

Chatbot Arena ranks language models with a Bradley–Terry model fit to pairwise human votes and shows each model's *rank spread*, the ranks consistent with per-model 95% rating intervals. Neither the 2024 intervals nor the current site's intervals carry a multiplicity correction. In a semi-synthetic simulation on Arena's real comparison graph, all 129 rank spreads hold simultaneously in 90.9% of datasets, below the 95% target. All six corrected procedures we study reach 95% simultaneous coverage in this simulation. Pairwise z-tests with Holm's step-down procedure give the narrowest corrected rank intervals in the simulation, 11.3 ranks on average, compared with 13.6 for Bonferroni intervals and 33–43 for the χ^2 set of the original Arena paper. Holm's directional control is proven only for exact ties; for near ties we checked it by simulation. On the category leaderboards, Higher Criticism, max-$|z|$ and χ^2 tests all reject the null of no specialties. Bonferroni flags 275 of 958 (model, category) pairs, but this count depends on how the null's noise level is calibrated.

1 Introduction

Chatbot Arena, now run by LMArena at arena.ai, is a widely used leaderboard for large language models. Users vote for the better of two anonymous answers, and the site ranks the models by a Bradley–Terry (BT) fit (Bradley and Terry, 1952) to these votes. Since 2025 it has also reported each model's *rank spread*, the best and worst rank consistent with all models' rating intervals (LMArena, 2025b), shown next to each rating's confidence interval.

The paper behind the site (Chiang et al., 2024) notes that a ranking depends on all ratings at once and therefore needs intervals that hold simultaneously, and it proposes a χ^2 confidence set for this purpose. The current open-source code (LMArena, 2025a) instead computes per-model 95% intervals, $\hat{r}_m \pm 1.96\,\mathrm{SE}_m$. Two features of this method affect coverage in opposite directions. Without a multiplicity correction, about 5% of the intervals are expected to fail to cover (about 6 of 129 models in 2024 and 20 of today's roughly 400). But the rank spread separates two models only when their intervals do not overlap, which for two independent estimates with equal standard errors is a test at $|z| > 1.96\sqrt{2} = 2.77$, the 99.4% level. We know of no study that measures their combined effect on simultaneous coverage. Arena also publishes category leaderboards (coding, math, Chinese and others). We call a *specialty* a (model, category) pair in which the model is stronger or weaker than its overall rating predicts; with about a thousand such pairs, some apparent specialties will arise by chance.

We study three questions. Q1 asks how often Arena's rank spreads contain the true ranks, both for individual models and for all models at once. Q2 compares procedures that correct for multiple testing, asking which reach 95% simultaneous coverage and how much width they add. Q3 asks whether category specialties exist and which model-category pairs show them. Because true ranks are unknown on real data, Q1 and Q2 use a semi-synthetic simulation in which the BT fit on the real 2024 battles is the truth

and only the outcomes are redrawn. For Q3 we calibrate specialty statistics by a parametric bootstrap and compare Higher Criticism (Donoho and Jin, 2004) with the maximum-$|z|$ (min-P) and χ^2 tests under the sparse-signal (Rare/Weak) model (Donoho and Jin, 2015).

Related work on inference for ranks includes bootstrap intervals (Hall and Miller, 2009), joint confidence regions for a ranking (Klein et al., 2020), and simultaneous confidence sets built from pairwise differences with step-down testing (Mogstad et al., 2024); our pairwise Holm procedure is the Bonferroni–Holm version of the last.

2 Data and Arena's method

Data

We use the public battle dump (LMSYS, 2024b) released with Chiang et al. (2024) and compare our implementation with the official leaderboard snapshot `elo_results_20240813` (last updated 2024-08-12; LMSYS, 2024a). The official filter (anonymous battles, the de-duplicated subsample, battles up to the snapshot time) leaves 1,657,517 battles among $M = 129$ models. Only 3,457 of the $K = \binom{129}{2} = 8{,}256$ pairs ever met, and 34.5% of battles are ties. Unlike the 2025 sample named in our proposal, this dump comes with an official reference fit computed from the same battle log, of which the dump is a slightly incomplete release (Appendix B). For the current site we use the 2026 boards in `leaderboard-dataset` (downloaded 2026-10-28; LMArena, 2026) and our record of the top 59 models that arena.ai displayed on 2026-10-08.

Model

Code a battle between models i and j as $y \in \{1, \frac{1}{2}, 0\}$ (win, tie, loss for i) and write $\sigma(x) = 1/(1 + e^{-x})$. Arena's `arena-rank` package estimates the strengths β by minimizing the logistic loss with fractional outcomes,

$$\hat{\beta} = \arg\min_{\beta} \; - \sum_{\text{battles}} \big[\, y\,(\beta_i - \beta_j) - \log\!\big(1 + e^{\beta_i - \beta_j}\big)\big], \tag{1}$$

whose population minimizer satisfies $\mathbb{E}[y] = \sigma(\beta_i - \beta_j)$. It reports ratings $\hat{r}_m = 1000 + (400/\ln 10)\,\hat{\beta}_m$. The covariance of $\hat{\beta}$ is estimated by the sandwich $\hat{V} = \hat{H}^{-1}\hat{G}\hat{H}^{-1}$ (White, 1980), with $\hat{H}$ the Hessian of (1) and $\hat{G}$ the covariance of the per-battle gradients; below, $\hat{V}$ is on the rating scale. Arena's rating intervals are $[L_m, U_m] = \hat{r}_m \pm 1.96\,\mathrm{SE}_m$ with $\mathrm{SE}_m = \hat{V}_{mm}^{1/2}$, and the rank spread is

$$\mathrm{best}_m = 1 + \#\{m' : L_{m'} > U_m\}, \qquad \mathrm{worst}_m = 1 + \#\{m' \neq m : U_{m'} > L_m\}. \tag{2}$$

We reserve *rank spread* for this construction and call the output of any procedure a *rank interval*. Its width is $\mathrm{worst}_m - \mathrm{best}_m + 1$, so a model with a single rank has width 1.

Validation against the published boards

Refitting the public battles with `arena-rank` reproduces the 2024 ratings to a median of 0.06 points, and the 2024 intervals have a half-width of 1.90 bootstrap SDs, as per-model 95% intervals should (Bonferroni would need 3.55). Every interval on the 4,773 published 2026 text-board snapshots is $\hat{r} \pm 1.9600\,\mathrm{SE}$, and (2) reproduces the site's rank spreads for all 59 recorded models (Appendix B). In 2024 as now, the published intervals carry no multiplicity correction, and the 2024 ranks match (2) up to a constant offset (Appendix B).

3 Methods

3.1 Semi-synthetic simulation

We fit (1) once on the real battles, treat the fitted strengths β^* as the truth, and define the true ranks $R_m = 1 + \#\{m' : \beta^*_{m'} > \beta^*_m\}$ (tied models share the best rank of their group). Each simulated dataset keeps the same 3,457 pairs with their real battle counts and redraws the outcomes. With $p = \sigma(\beta^*_i - \beta^*_j)$ and the observed tie rate $t = 0.345$, a battle is a win, tie or loss for i with probabilities $p - t/2$, t and $1 - p - t/2$, so that $\mathbb{E}[y] = p$ and β^* is exactly the estimand of (1). We refit, apply every procedure to the same fit, and repeat 1,000 times. We record per-model and simultaneous coverage, mean width, and *false orderings* ("i above j" declared although $\beta^*_i \le \beta^*_j$), each with a Monte Carlo SE (Appendix A).

Lemma 1. *Let a procedure declare a set D of orderings "i above j", and let $\text{best}_m = 1 + \#\{i : (i,m) \in D\}$ and $\text{worst}_m = M - \#\{j : (m,j) \in D\}$. For $D = \{(i,j) : L_i > U_j\}$ this is the rank spread* (2)*. If D contains no false ordering, then $\text{best}_m \le R_m \le \text{worst}_m$ for every m.*

Proof. If all declared orderings are true, the models declared above m are among the $R_m - 1$ models truly above it, and those declared below m are among the at most $M - R_m$ models truly below it. □

Hence any procedure whose probability of a false ordering is at most 5% has simultaneous rank coverage of at least 95% (cf. Mogstad et al., 2024). False orderings are most likely when two models have nearly equal strengths.

3.2 Corrected procedures (Q2)

Table 1 lists when each procedure declares i above j; Lemma 1 turns these orderings into rank intervals. Bonferroni and the χ^2 set widen the intervals and keep the overlap rule. The χ^2 set of Chiang et al. (2024),

$$\{\beta : (\hat\beta - \beta)^\top \hat V^- (\hat\beta - \beta) \le \chi^2_{M-1,\,0.95}\},$$

is used either projected onto each coordinate or through pairwise contrasts (Scheffé, 1953). The pairwise procedures test each pair directly with

$$z_{ij} = \frac{\hat r_i - \hat r_j}{\text{SE}(\hat r_i - \hat r_j)}, \qquad \text{SE}(\hat r_i - \hat r_j)^2 = \hat V_{ii} + \hat V_{jj} - 2\hat V_{ij}.$$

In these units the overlap rule requires $|z_{ij}| > 1.96\,\rho_{ij}$ with $\rho_{ij} = (\text{SE}_i + \text{SE}_j)/\text{SE}(\hat r_i - \hat r_j)$. On the real fit ρ_{ij} ranges from 1.09 to 1.73 (median 1.37); it exceeds $\sqrt{2}$ only for the 20% of pairs whose estimates are clearly positively correlated. *Pairwise Bonferroni* uses the critical value $z_{1-\alpha/(2K)} = 4.52$ for all pairs. *Holm*'s procedure (Holm, 1979) compares the k-th largest $|z|$ with $c_k = z_{1-\alpha/(2(K-k+1))}$, stops at the first pair that fails, and declares each rejected pair in the direction of z_{ij}. *One-sided Holm* applies Holm's procedure to the $2K$ hypotheses $H_{ij} : \beta_i \le \beta_j$, whose rejection declares the ordering "i above j"; its k-th critical value is $z_{1-\alpha/(2K-k+1)}$.

Guarantees

By Lemma 1 the quantity to control is the probability of any false ordering. Bonferroni-type rules bound it by the union bound. Holm's theorem bounds the probability of rejecting any *true* hypothesis, and for one-sided Holm the true hypotheses are exactly the false orderings. For Holm, an exactly tied pair is a true null and is covered by the theorem, but a pair that is close and not tied is a false null, and its

Table 1: When each procedure declares model i above model j (gap $= \hat{r}_i - \hat{r}_j$; $M = 129$, $K = 8{,}256$, $\alpha = 0.05$). "Proven": the probability of any false ordering is at most α under any dependence, up to the normal approximation.

Procedure	Rule	False orderings controlled
Arena (uncorrected)	gap $> 1.96\,(\mathrm{SE}_i + \mathrm{SE}_j)$	no
Bonferroni intervals	gap $> 3.55\,(\mathrm{SE}_i + \mathrm{SE}_j)$	proven
χ^2 projection	gap $> 12.47\,(\mathrm{SE}_i + \mathrm{SE}_j)$	proven
χ^2 Scheffé	$z_{ij} > 12.47$	proven
Pairwise Bonferroni	$z_{ij} > 4.52$	proven
Holm	$z_{ij} > c_k$, step-down from 4.52 toward 1.96	for exact ties only
One-sided Holm	$z_{ij} > c_k$, step-down from 4.52, never below 4.38	proven

possible error is a wrong *direction*. Directional control after a two-sided Holm procedure is proven for independent normal statistics (Shaffer, 1980), but our z_{ij} are dependent because pairs share models, and for dependent statistics only special structures are covered (Finner, 1999; Guo and Romano, 2015); we test it by simulation. The proven one-sided version gains little from stepping down: at most one of H_{ij} and H_{ji} can be rejected, so at step $k \le K$ at least $K+1$ hypotheses remain and its critical value never falls below $z_{1-\alpha/(K+1)} = 4.38$. The declared orderings of all three pairwise procedures are always transitive (Appendix D).

Benchmark and stress tests

As an oracle benchmark we recompute rank spreads from $\hat{\beta} \pm z\,\mathrm{SE}$ on a grid of z from 1.8 to 4 and find the smallest z that reaches 95% simultaneous coverage; it uses the true ranks, so it is not a procedure. To test the guarantees where they bind, we rerun the simulation on the same graph with all strengths scaled by 0.5, 0.25, 0.1 or 0; with the models sorted into 20 or 10 equal-size tiers of exactly equal strength (354 or 768 tied pairs); and with 20 tiers in which each model keeps 5%, 10% or 20% of its distance from the tier mean (median within-tier gaps of 0.1–0.4 SEs), where every false ordering is a wrong-direction error not covered by Holm's theorem.

3.3 Category specialties (Q3)

We define eight categories from the dump's prompt tags (coding, math, instruction following, hard prompts, Chinese, Russian, multi-turn, long query). For models with enough battles, each definition matches the official 2024 board of the same name (Appendix E). We test the $N = 958$ (model, category) pairs with at least 200 battles in the category. For model m in category c, let y_b be its score in battle b and $p_b = \sigma(\hat{\beta}_m - \hat{\beta}_{o(b)})$ the win probability implied by the *overall* fit against opponent $o(b)$. The specialty statistic is

$$z_{mc} = \textstyle\sum_b (y_b - p_b) \Big/ \sqrt{\sum_b (y_b - p_b)^2},$$

summed over m's battles in c (the empirical variance accounts for ties). The overall fit uses these battles too, so z_{mc} is shrunk toward zero and the z_{mc} are correlated. We therefore calibrate by a parametric bootstrap under the null of no specialties: redraw all outcomes from the overall fit and refit, 500 times. Under the null the raw z_{mc} have standard deviations from 0.70 to 1.05. Each z_{mc} is standardized against its own null mean and SD to z^*_{mc}, with two-sided p-values from $N(0,1)$.

To test whether *any* specialty exists we use Higher Criticism,

$$\mathrm{HC} = \max_{k \le N/2} \frac{\sqrt{N}\,(k/N - p_{(k)})}{\sqrt{p_{(k)}(1 - p_{(k)})}},$$

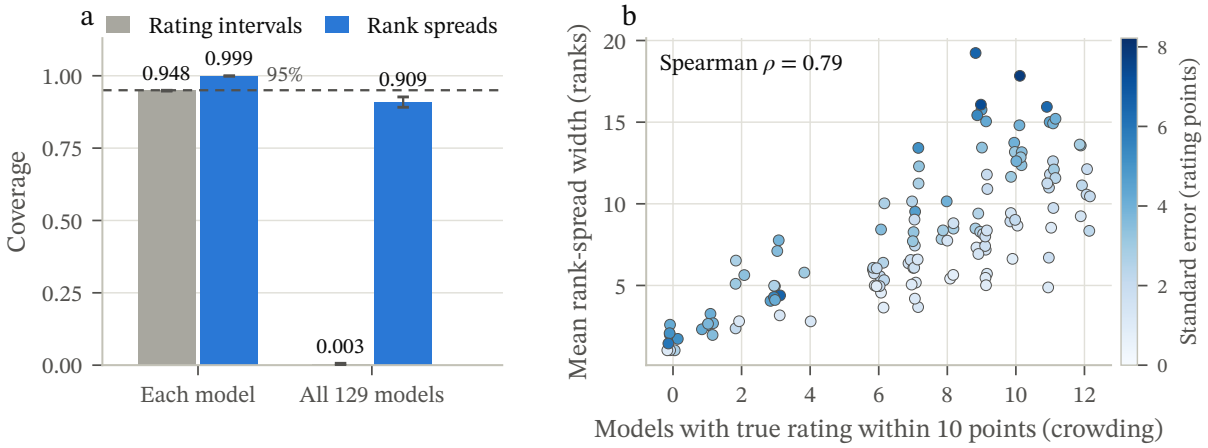


Figure 1: Q1, 1,000 simulated datasets. (a) Coverage of Arena's rating intervals and rank spreads, for each model and for all 129 models at once, with 95% Monte Carlo intervals. (b) Mean rank-spread width of each model against the number of models whose true rating is within 10 points of its own (horizontally jittered), colored by the model's standard error.

its variant HC$^+$ restricted to $p_{(k)} > 1/N$, the maximum-$|z^*|$ test (min-P) and $\chi^2 = \sum z^{*2}$, each with its p-value from the null draws. To find *which* pairs, we use Bonferroni at nominal family-wise level 5% and Benjamini–Hochberg (Benjamini and Hochberg, 1995) at nominal false discovery rate 10%. For power, we plant $s = N^{1-\vartheta}$ specialties with random signs at random pairs, for $\vartheta \in \{0.2, 0.3, \dots, 0.7\}$, with strength set so that each planted z has mean $\mu = \sqrt{2r \log N}$ before the refit, for $r \in \{0.1, 0.2, 0.4, 0.8, 1.5\}$; 200 replicates per cell, with critical values from fresh null draws.

4 Results

4.1 Q1: Coverage of Arena's rank spreads

Each rating interval covers its true rating 94.8% of the time, but on average 6.65 of the 129 fail to cover, and all 129 cover in only 3 of 1,000 datasets (Figure 1a). The rank spreads cover far more often: each contains its true rank 99.9% of the time, and all 129 do so in 90.9% of datasets (Monte Carlo SE 0.9%). The shortfall from 95% is small but well beyond Monte Carlo error (4.5 SEs).

By Lemma 1 a rank spread fails only through a false ordering, and the overlap rule's conservatism absorbs most of the rating intervals that fail to cover. The rank spread is very conservative for each model and slightly anti-conservative jointly. Its simultaneous coverage barely changes when all strengths are scaled down (90.1–91.5% at scales 0.5 to 0.1, which act like up to 100 times less data), but it falls sharply once models are exactly or nearly tied (Section 4.2). Across models, spread width is most strongly associated with crowding, the number of models within 10 rating points (Spearman correlation 0.79; Figure 1b). Given crowding, the standard error also matters (OLS $t = 8.9$), but the true rank does not ($t = 1.1$; Appendix C).

4.2 Q2: Corrected rank intervals

Every correction covers all true ranks in all 1,000 baseline datasets, with no false orderings observed. Their mean widths differ by almost a factor of four (Figure 2a). The χ^2 set gives the widest intervals: 33.1 ranks read through contrasts and 43.4 projected. It protects every linear combination of the 129 strengths, and its critical value $(\chi^2_{128,\,0.95})^{1/2} = 12.47$ grows like $\sqrt{M}$, whereas a ranking depends only on the signs of K differences, whose Bonferroni critical value grows only like $\sqrt{2 \log K}$. Bonferroni intervals need 13.6

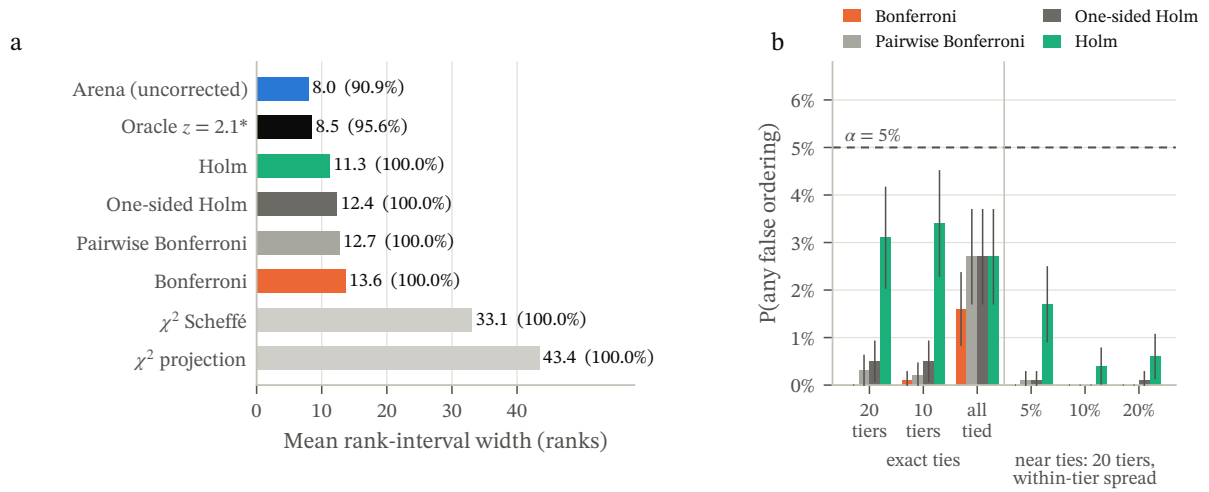


Figure 2: Q2. (a) Mean rank-interval width on the 1,000 datasets of Q1; in parentheses, the share of datasets in which all 129 rank intervals contain the true ranks. *The oracle chooses z using the true ranks. (b) Probability of any false ordering in the stress-test worlds (1,000 datasets each), with 95% Monte Carlo intervals; the dashed line is the 5% target. Arena is not shown in (b) because its values (34–100%) are off the scale.

ranks. Holm needs 11.3, 18% less, and it detects 92.0% of the true orderings, compared with 90.1% for Bonferroni.

Both direct pairwise testing and the step-down procedure reduce width. Testing each difference directly saves 0.9 ranks (13.6 to 12.7), because the overlap rule effectively multiplies the critical value by $\rho_{ij} \approx 1.37$ ($3.55 \times 1.37 \approx 4.86 > 4.52$). Holm's step-down saves 1.5 ranks more: about 7,600 pairs are rejected, so only about 660 remain, and the critical value falls from 4.52 to $z_{1-\alpha/(2\cdot 660)} = 3.96$. One-sided Holm reaches only 12.4 ranks, because its critical value cannot fall below 4.38. The oracle reaches 95.6% coverage at $z = 2.1$ with 8.5 ranks.

Stress tests

A pair whose true gap is δ standard errors is declared in the wrong direction with probability about $\Phi(-\delta - c)$, where c is its critical value, so errors come from near-ties, which the 2024 strengths rarely have. Figure 2b shows the worlds where family-wise bounds bind. With 20 or 10 exactly tied tiers, Holm's critical value falls to 3.96 or 4.01 and Holm declares a false ordering in 3.1% and 3.4% of datasets (Monte Carlo SE 0.5%), close to the union-bound approximations $354 \times 2\Phi(-3.96) \approx 2.7\%$ and $768 \times 2\Phi(-4.01) \approx 4.6\%$; the other pairwise procedures stay at or below 0.5%. When all models are tied, the three pairwise procedures coincide, with a false-ordering rate of 2.7%.

In the near-tie worlds, where Holm has no proven guarantee, its error rate is 1.7% at a 5% spread and 0.4–0.6% at larger spreads, with simultaneous coverage of at least 99.4%. Exact ties, which Holm's theorem covers, are thus the harder case in our simulations, and there Holm is close to the oracle: the oracle needs 10.7 ranks with 20 tied tiers (Holm 11.3) and 13.8 with 10 tiers (Holm 14.0). Arena's uncorrected spreads all hold in only 24%, 7% and 0.2% of datasets with 20 tiers, 10 tiers and all models tied (Appendix D).

The real 2024 board

Holm keeps the top three models at single ranks, as Arena does, whereas Bonferroni ties the top two (Figure A2); mean widths are 7.9 ranks for Arena, 11.4 for Holm and 13.8 for Bonferroni. Of the 7,808 orderings Arena declares, Holm does not support 223, whose $|z|$ lie between 2.5 and 3.96: each is significant at the 5% level by itself, and none passes the correction for 8,256 comparisons. Conversely, Holm separates none of the 448 pairs whose rating intervals overlap, because Arena's implicit critical value $1.96\,\rho_{ij}$ is at most 3.4. Holm's rank intervals also separate none of the 830 pairs whose rank spreads

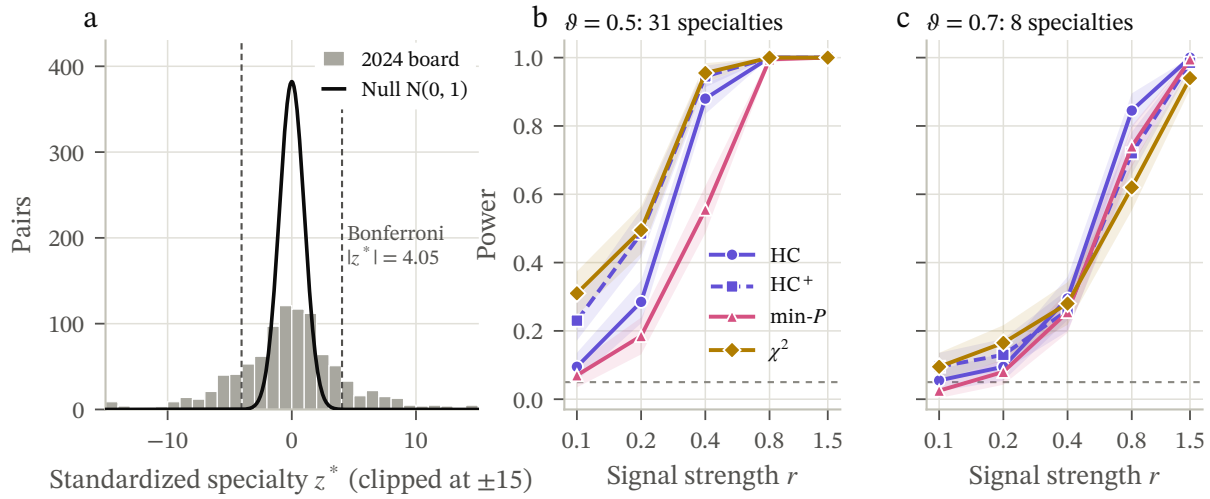


Figure 3: Q3. (a) Standardized specialty statistics z^* of the 958 tested pairs on the 2024 board against the null $N(0, 1)$; dashed lines mark the Bonferroni critical value. (b, c) Power of the four global tests in simulated data with 31 ($\vartheta = 0.5$) and 8 ($\vartheta = 0.7$) planted specialties, 200 replicates per point; shaded bands are 95% Monte Carlo intervals and the dotted line is the 5% level.

Table 2: Q3 on the 2024 board ($N = 958$ tested pairs). The null distribution comes from 500 parametric-bootstrap draws, so the smallest attainable p-value is $1/501$. Plain HC is dominated by the smallest p-value ($\log_{10} p \approx -353.6$); we compute it from log normal tails to avoid underflow. HC$^+$ uses only $p > 1/N$ and excludes these extreme tails.

Test	Observed	Null 95% point	p
HC$^+$	330	3.1	1/501
χ^2	21,210	1,028	1/501
min-P (max $\lvert z^* \rvert$)	40.3	4.05	1/501
HC	1.98×10^{175}	4.4	1/501

overlap (the site's tie rule). This second result is only empirical.

4.3 Q3: Category specialties

The 2024 category effects depart substantially from the null (Figure 3a, Table 2). Every global test rejects at the smallest attainable p-value. A naive cut at $p < 0.05$ would flag 521 pairs compared with 48 expected under the null. Bonferroni flags 275 pairs, in every category (17–40% of the pairs tested in each) and for 94 of the 129 models, and the median |effect| of the flagged pairs is 25 rating points. BH, at its 10% level, flags 525, slightly more than the naive cut, because its cutoff $0.1 \times 525/958$ exceeds 0.05. The largest effects concern language: `llama-3-70b-instruct` is 73 points weaker in Chinese than overall ($z^* = -40$), and all nine Qwen models are 50–84 points stronger in Chinese than overall (eight pass Bonferroni, all nine pass BH).

The evidence that specialties exist is stronger than the evidence for any particular count. The null does not account for correlated votes (same user or session, clusters in time) and any misfit of the BT model, and the robust SD of the observed z^* is 3.3 rather than 1 (an overstatement of the noise, since it includes the true specialties). If the true noise were 1.5, 2 or 3 times the null's, Bonferroni would flag 149, 76 or 26 pairs instead of 275; we have not estimated this factor.

The power simulation (Figure 3b,c) reproduces the qualitative predictions of the Rare/Weak model, in which $s = N^{1-\vartheta}$ signals of size μ are detectable only above a boundary in (ϑ, r). With 31 planted specialties ($\vartheta = 0.5$) at $r = 0.4$, HC and HC$^+$ have power 0.88 and 0.95 and min-P only 0.56, and with 122 ($\vartheta = 0.3$)

at $r = 0.2$ min-P reaches only 0.57 while χ^2 and HC$^+$ reach 1.00. With 8 ($\vartheta = 0.7$) χ^2 falls behind, as the theory predicts: at $r = 0.8$ its power is 0.62, min-P 0.74, HC 0.85 and HC$^+$ 0.72. HC leads min-P by about four Monte Carlo SEs, but HC$^+$, the form used on the real board, does not beat min-P here, and at $\vartheta = 0.7$ the theoretical boundaries of HC and the max test nearly coincide. For identification, BH finds 76–93% of the strongest planted specialties ($r = 1.5$, about 27 rating points at the median pair) and Bonferroni about 53%; in between lies a wide region where χ^2 and HC$^+$ detect specialties that neither Bonferroni nor BH can locate. On the real board, all global statistics exceed their null 95% points by a wide margin, well beyond this detection-only region.

5 Discussion

Arena's rank spreads fall short of the simultaneous target and fail badly when models are tied; in our simulations Holm meets the target with the narrowest intervals of the corrections. Category specialties are widespread, but their number depends on the null's noise level.

Interpretation

By Lemma 1, rank coverage only needs the declared orderings to be error-free. The χ^2 set controls every linear combination of the strengths, most of which cannot change a ranking. The overlap rule amounts to a test at $|z| > 1.96\,\rho_{ij}$ with ρ_{ij} near 1.4. Arena's uncorrected spreads therefore come close to the target in the baseline simulation, and no correction separates a pair whose rating intervals overlap. The error rate of a pairwise procedure is roughly the number of nearly tied pairs times the tail probability beyond their critical value. That critical value drops only as well-separated pairs are rejected, so tiers of tied models among well-separated ones are the worst case. Resampling the joint distribution of the z_{ij} (Romano and Wolf, 2005; Mogstad et al., 2024) or using the logical constraints among pairwise hypotheses (Shaffer, 1986) can only lower Holm's critical values; on the real board the first adds 18 orderings.

Recommendation

If Arena wants its rank spreads to mean that the true ranks lie inside jointly with 95% probability, pairwise tests with Holm are a simple replacement, which met that target in all our simulations. They need only the covariance matrix that `arena-rank` already computes, on the real 2024 board they widen the rank spreads by 3.5 ranks on average and keep the top three models at single ranks; if a proven guarantee is required, one-sided Holm gives intervals 1.1 ranks wider. If the current spreads are kept, the site should state that they are not joint statements; on its own, each one held in 99.9% of our idealized simulations.

Limitations and future work

The simulation assumes independent battles, an exact BT model, fixed strengths and one tie rate; correlated votes or drifting strengths would likely lower real coverage. The true strengths are estimates, slightly more spread out than the real ones, which makes coverage easier in the simulation. The simulation uses one world of 129 models with 2024 strengths, whereas today's board has about 400 models, where pairwise Bonferroni would need $z = 4.98$. All guarantees rely on the normal approximation, and Holm's directional control rests on simulation. Q3's null ignores correlated votes and anything else that varies by category, such as drift combined with a changing category mix. The largest specialties are language effects, where the voters also differ: Chinese prompts are judged by different users, so a language specialty may measure voter preferences as much as model ability. A bootstrap over users or days would calibrate Q3's noise level and Q1's coverage under correlated votes, and synthetic worlds with 200–400 models would show how the conclusions scale. Appendix F lists further limitations.

References

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.

Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Banghua Zhu, Hao Zhang, Michael I. Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot arena: An open platform for evaluating LLMs by human preference. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2403.04132.

David Donoho and Jiashun Jin. Higher criticism for detecting sparse heterogeneous mixtures. *The Annals of Statistics*, 32(3): 962–994, 2004.

David Donoho and Jiashun Jin. Higher criticism for large-scale inference, especially for rare and weak effects. *Statistical Science*, 30(1):1–25, 2015.

Bradley Efron. Large-scale simultaneous hypothesis testing: The choice of a null hypothesis. *Journal of the American Statistical Association*, 99(465):96–104, 2004.

Helmut Finner. Stepwise multiple test procedures and control of directional errors. *The Annals of Statistics*, 27(1):274–289, 1999.

Wenge Guo and Joseph P. Romano. On stepwise control of directional errors under independence and some dependence. *Journal of Statistical Planning and Inference*, 163:21–33, 2015. doi: 10.1016/j.jspi.2015.02.009.

Peter Hall and Hugh Miller. Using the bootstrap to quantify the authority of an empirical ranking. *The Annals of Statistics*, 37 (6B):3929–3959, 2009.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

Martin Klein, Tommy Wright, and Jerzy Wieczorek. A joint confidence region for an overall ranking of populations. *Journal of the Royal Statistical Society: Series C*, 69(3):589–606, 2020.

LMArena. Arena-rank: Open-sourcing the leaderboard methodology. Blog post and code, https://github.com/lmarena/arena-rank, 2025a.

LMArena. Arena's ranking method. Blog post, https://arena.ai/blog/ranking-method. Updated February 2026, 2025b.

LMArena. `leaderboard-dataset`. Hugging Face, https://huggingface.co/datasets/lmarena-ai/leaderboard-dataset, 2026.

LMSYS. Chatbot arena leaderboard snapshot `elo_results_20240813.pkl`. Hugging Face Space, https://huggingface.co/spaces/lmarena-ai/chatbot-arena-leaderboard, 2024a.

LMSYS. Public Chatbot Arena battle data, `clean_battle_20240814_public.json`. https://storage.googleapis.com/arena_external_data/public/clean_battle_20240814_public.json, 2024b.

Magne Mogstad, Joseph P. Romano, Azeem M. Shaikh, and Daniel Wilhelm. Inference for ranks with applications to mobility across neighbourhoods and academic achievement across countries. *The Review of Economic Studies*, 91(1):476–518, 2024.

Joseph P. Romano and Michael Wolf. Exact and approximate stepdown methods for multiple hypothesis testing. *Journal of the American Statistical Association*, 100(469):94–108, 2005.

Henry Scheffé. A method for judging all contrasts in the analysis of variance. *Biometrika*, 40(1/2):87–104, 1953.

Juliet Popper Shaffer. Control of directional errors with stagewise multiple test procedures. *The Annals of Statistics*, 8(6): 1342–1347, 1980.

Juliet Popper Shaffer. Modified sequentially rejective multiple test procedures. *Journal of the American Statistical Association*, 81(395):826–831, 1986.

Halbert White. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4):817–838, 1980.

Appendix

A Implementation details and reproducibility

Code and data

All code, results and figures are at https://github.com/chenxige0509/stat236-arena-uncertainty. The README lists one command per step, and each script states its inputs, outputs and the numbers it establishes. The battle dump (2.1 GB) is not in the repository; `prepare_data.py` converts it to Parquet. The official snapshot is downloaded automatically and read with an allow-list unpickler.

Estimator

We use `arena-rank` 0.1.1 with its defaults. `BradleyTerry` minimizes (1) by L-BFGS from zero (at most 1,000 iterations, tolerances 10^{-9}). The Hessian has a ridge of 10^{-5} times the number of battles on its diagonal, which also fixes the location of the strengths, and the intervals use the sandwich covariance with $z = 1.96$. The package returns only the diagonal of $\hat{V}$; we recompute the full matrix from the same Hessian and gradient covariance. Other software: Python 3.12–3.13, JAX 0.8.1, NumPy 2.3.3, pandas 3.0.6, SciPy 1.18.1, statsmodels 0.15.0.

Simulation

Each dataset is drawn as multinomial win/tie/loss counts per pair, which is equivalent to drawing the 1.66 million battles one by one, and stored in a fixed-shape array, so the BT fit compiles once and one fit takes about 0.2 seconds. The tie probability is capped at $2\min(p, 1-p)$ so that the probabilities stay valid; the cap binds only for rating gaps above about 270 points and keeps $\mathbb{E}[y] = p$. A test (`test_simulate.py`) checks that this format reproduces `arena-rank`'s own data loader to 10^{-9} and that simulated estimates are unbiased. Q1 and Q2 use seed 0 and identical draws, and `q2.py` asserts that its Arena rank spreads equal those of Q1. Each Monte Carlo SE is the standard deviation of the per-dataset quantity divided by $\sqrt{1000}$ (for a proportion $\hat{\pi}$, $\sqrt{\hat{\pi}(1-\hat{\pi})/1000}$). The oracle grid has step 0.05 from 1.8 to 4.0, plus 1.96 and 3.548. The pairwise procedures work on $|z|$ and normal quantiles rather than p-values, because thousands of p-values underflow; `test_pairwise.py` checks them against `statsmodels` on 200 random correlated problems. Q3 uses seed 0 for the 500 null draws and seed 1 for the power grid, whose critical values and size come from two further sets of 400 fresh null draws. A planted specialty δ_{mc} shifts model m's strength in every battle of category c by $\mu\sqrt{\mathrm{Var}_{mc}}/I_{mc}$, where $I_{mc} = \sum p(1-p)$ and $\mathrm{Var}_{mc} = \sum[p(1-p) - t/4]$ over the pair's battles; at $r = 1.5$ the realized mean $|z^*|$ after the refit is 0.91μ. Runtimes on a laptop (Apple silicon): Q1 3 minutes, Q2 4 minutes per world, Q3's calibration 2 minutes, Q3's power grid 35 minutes.

B Validation details

Table A1: Checks of our implementation against Arena's published numbers.

Check	Result
2024 battles per model	91 of 128 models within ±10 battles; recent models miss 1,000–2,000 battles absent from the public dump
2024 ratings	median absolute difference 0.06 points; at most 0.3 for models whose battle counts match
2024 intervals	identical to the bootstrap quantiles; half-width / SD = 1.90 (per-model level: 1.88; Bonferroni: 3.55; χ^2: 12.42)
2026 intervals (4,773 snapshots)	half-width / $\sqrt{\text{variance}} = 1.9600$ for all 1.5 million model rows (Bonferroni would need 2.24–3.85)
2026 rank spreads	formula (2) reproduces both ends for 59 of 59 models

2024. (1) With the official filter, 30 models match the official battle count exactly and 91 of 128 within

±10; without the de-duplication filter no model matches within ±10. (2) After shifting our ratings so that `mixtral-8x7b-instruct-v0.1` has the board's anchor value 1114, the median absolute difference is 0.06 points (correlation 0.99999). (3) The published intervals equal the 2.5% and 97.5% quantiles of the 100 stored bootstrap fits exactly; their half-width is 1.90 bootstrap SDs (median), against 1.88 expected for a per-model 95% interval estimated from 100 normal draws. Our sandwich SE is 0.98 times the bootstrap SD (median), so the current sandwich intervals and the 2024 bootstrap intervals have the same width. (4) The published rank equals $1 + \#\{m' : L_{m'} > U_m\}$ up to a constant offset of 5 for 112 models (0–4 for the top 16), consistent with the board having been ranked while five unreleased models were listed. Our fit has 129 models, one more than the board, because the dump contains one of the board's hidden pre-release models.

2026. The site's default text view is the style-controlled board in the `leaderboard-dataset` release: on 2026-10-08 all 59 recorded scores agree with it to rounding (at most 0.50), as do all ± values and vote counts, while the board without style control differs by up to 54 points. On all 4,773 text-board snapshots published in 2026 (as downloaded on 2026-10-28; the dataset keeps growing), the ratio of half-width to $\sqrt{\text{variance}}$ is 1.9600 to four decimals. A Bonferroni or χ^2 correction would need 2.24–3.85 or 2.0–21.6 depending on the board's size. Only the text boards and one site view were checked.

C Q1: supplementary results

Table A2: Q1 by true-rank tier (means over the models in each tier and over 1,000 datasets).

True rank	Battles	SE	Models within 10 points	Rank exactly right	Spread width
1–10	37,376	1.88	4.1	77%	3.7
11–30	50,332	1.99	6.9	48%	6.1
31–60	35,178	1.96	7.8	43%	6.8
61–100	17,152	2.99	9.8	30%	12.1
101–129	6,662	4.17	4.0	57%	6.4

Of the 103 rank spreads that failed in the simulation, 47% made the model look worse than it is (best rank numerically larger than the true rank) and 53% made it look better. Counting only orderings whose rank spreads do not overlap, the site's tie rule, a false ordering appeared in 0.6% of datasets and 90.0% of the true orderings were found; counting every ordering the interval rule declares, a false ordering appeared in 15.5% of datasets. When all strengths are scaled down, Arena's probability of a false ordering rises to 29.5%, 44.3% and 64.4% at scales 0.5, 0.25 and 0.1 while its simultaneous rank coverage stays near 91%. Lemma 1 is only a sufficient condition: a false ordering breaks coverage only when it pushes a model's count past its true rank, and under the site's tie rule the false-ordering rate stays at 0.1–0.6%. Spearman correlations with the spread width across the 129 models are 0.79 for crowding, 0.27 for the standard error, 0.23 for the true rank, −0.18 for the model's release date and 0.04 for the number of distinct opponents. In an OLS regression of the width on standardized drivers that includes crowding, the standard error has coefficient 2.62 ($t = 8.9$) and the true rank 0.29 ($t = 1.1$). These analyses are correlational, and the drivers are related to each other (older models are weaker and have fewer battles).

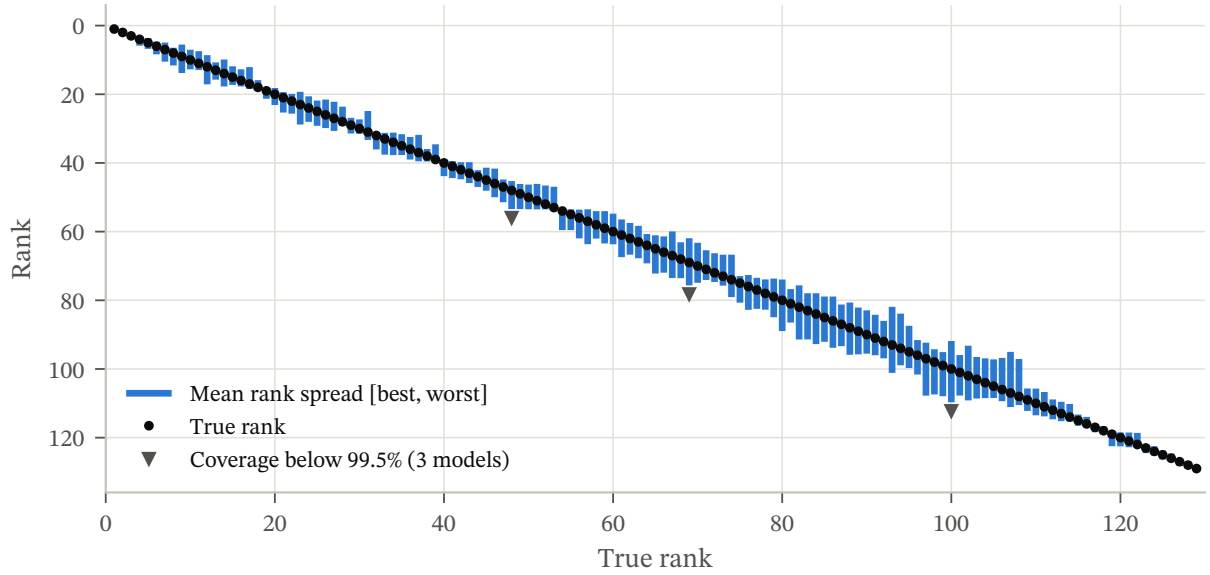


Figure A1: Mean rank spread of each model over the 1,000 simulated datasets, against its true rank. Triangles mark the three models whose rank spread contains the true rank in fewer than 99.5% of datasets.

D Q2: supplementary results

Table A3: Q2 on the 1,000 datasets of Q1. "Declared": orderings the procedure declares; "rank rule": pairs whose rank intervals do not overlap. Monte Carlo SEs are below 0.01 for coverage and widths.

	Coverage		Width	P(any false ordering)		True orderings
Procedure	each	all	(ranks)	declared	rank rule	declared
Arena	0.999	0.909	7.99	0.155	0.006	0.945
Bonferroni	1.000	1.000	13.65	0	0	0.901
χ^2 projection	1.000	1.000	43.43	0	0	0.669
χ^2 Scheffé	1.000	1.000	33.07	0	0	0.749
Pairwise Bonferroni	1.000	1.000	12.73	0	0	0.908
One-sided Holm	1.000	1.000	12.38	0	0	0.911
Holm	1.000	1.000	11.25	0	0	0.920

Bonferroni's rating intervals all cover in 94.5% of datasets (Monte Carlo SE 0.7%). The union bound is therefore nearly tight for the ratings. Converting those intervals to rank intervals adds conservatism.

Uniformly scaled-down strengths are not a hard case for the corrections: the gaps shrink, but so does the number of declared orderings, and few pairs end up close enough to be at risk. At a critical value near 4, a true gap of one standard error makes a wrong-direction declaration 100–200 times less likely than for an exactly tied pair.

Declared orderings of the pairwise procedures are transitive: each procedure declares exactly the pairs whose $|z|$ exceeds some cutoff, and because $\mathrm{SE}(\hat{r}_i - \hat{r}_k) \le \mathrm{SE}(\hat{r}_i - \hat{r}_j) + \mathrm{SE}(\hat{r}_j - \hat{r}_k)$, declaring $i > j$ and $j > k$ implies $z_{ik} \ge \min(z_{ij}, z_{jk})$. The rank intervals therefore come from a proper partial order, and `q2.py` checks this in every dataset.

The real board, pair by pair

Of the 8,256 pairs, Arena's interval rule orders 7,808 and Holm 7,585, all of which Arena also orders. Under the site's tie rule, Arena separates 7,426 pairs; for 428 of them the Holm rank intervals overlap, and none of the 830 pairs whose rank spreads overlap has separate Holm rank intervals. Holm declares every ordering that Bonferroni's intervals declare, and 155 more. A single-step max-T test, with its critical value set to the 95th percentile of

Table A4: Simultaneous coverage of the rank intervals in the stress-test worlds (1,000 datasets each). Monte Carlo SEs are at most 0.016 for Arena and 0.006 for the others.

World	Arena	Bonferroni	Pairwise Bonf.	One-sided Holm	Holm
2024 strengths	0.909	1.000	1.000	1.000	1.000
Strengths × 0.5	0.901	1.000	1.000	1.000	1.000
Strengths × 0.25	0.914	1.000	1.000	1.000	1.000
Strengths × 0.1	0.915	1.000	1.000	1.000	1.000
20 tiers, exactly tied	0.241	1.000	0.998	0.996	0.975
10 tiers, exactly tied	0.073	0.999	0.998	0.995	0.966
20 tiers, spread 5%	0.668	1.000	0.999	0.999	0.994
20 tiers, spread 10%	0.703	1.000	1.000	1.000	0.998
20 tiers, spread 20%	0.782	1.000	1.000	1.000	1.000
All models tied	0.002	0.984	0.973	0.973	0.973

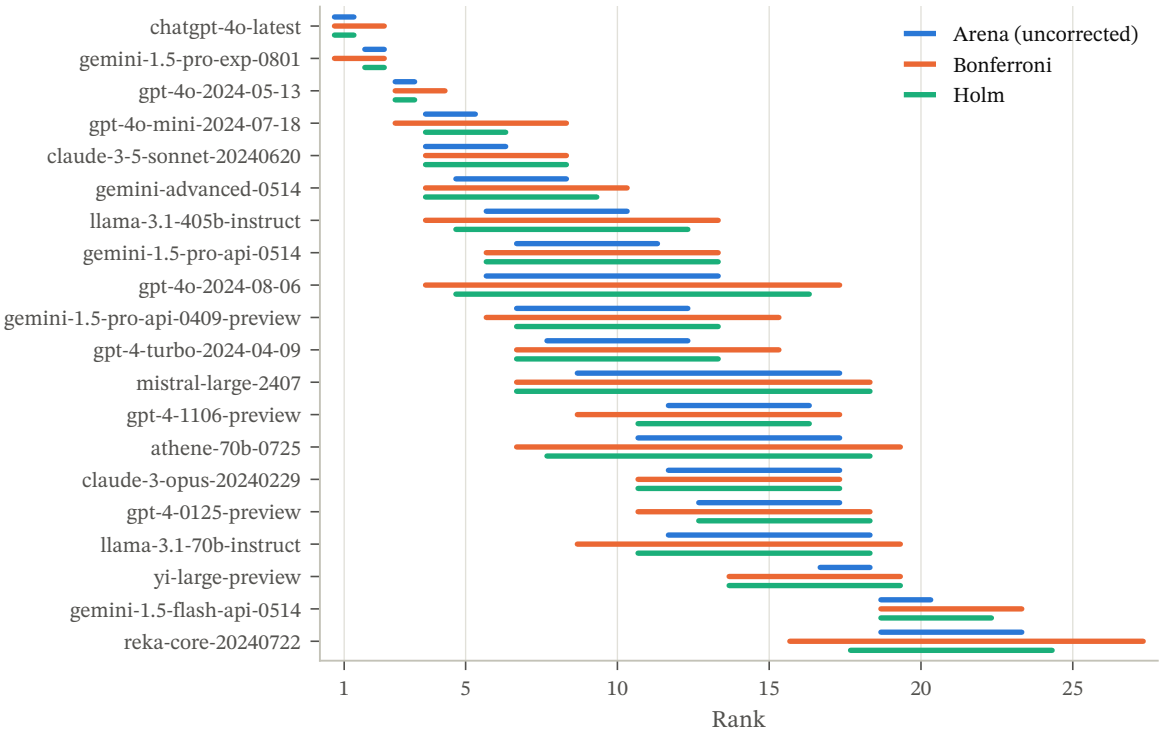


Figure A2: Rank intervals of the 20 highest-rated models on the 2024 board (our refit of the public battles) under Arena's method, Bonferroni intervals and Holm. Each model has one bar per procedure, offset vertically.

$\max_{i<j} |Z_i - Z_j|/\mathrm{SE}(\hat{r}_i - \hat{r}_j)$ for $Z \sim N(0, \hat{V})$ (20,000 draws), needs 4.30 instead of 4.52 and separates 7,534 pairs; its step-down version (Romano and Wolf, 2005) stops at 3.86 and separates 7,603, 18 more than Holm.

E Q3: supplementary results

For the tested pairs, our category-minus-overall ratings have correlation at least 0.999 with the official ones. Below 200 battles a few old models disagree with the official category boards by up to 408 points; the official values suggest that the fit may not have converged (three Russian models share an identical rating), which motivated the 200-battle minimum. The null 95% points in Table 2 use the same 500 draws that set the standardization; with leave-one-out standardization they are 1,036 for χ^2 and 4.12 for max $|z^*|$, and no p-value changes. The sizes of the global tests under the null, from fresh draws, are 0.06 (HC), 0.05 (HC$^+$), 0.05 (min-P) and 0.08 (χ^2); only χ^2 exceeds 5%, by 2.4 Monte Carlo SEs. The identification simulation records custom attribution diagnostics, which do not establish standard FDR/FWER control (Appendix F).

Table A5: Q3 categories. "Within ±10": models (of 128) whose battle count in the category is within 10 of the official category board's. "Tested": pairs with at least 200 category battles.

Category	Battle belongs when	Battles	Within ±10	Tested	Bonferroni
Coding	code detected	316,372	98	128	35
Math	math tag	214,348	98	126	40
Instruction following	instruction-following tag	581,492	97	129	42
Hard prompts	≥ 6 of 7 criteria tags	447,819	98	129	39
Chinese	language	176,016	100	115	46
Russian	language	143,973	99	95	32
Multi-turn	more than one turn	244,413	98	128	22
Long query	≥ 500 user tokens	164,230	99	108	19

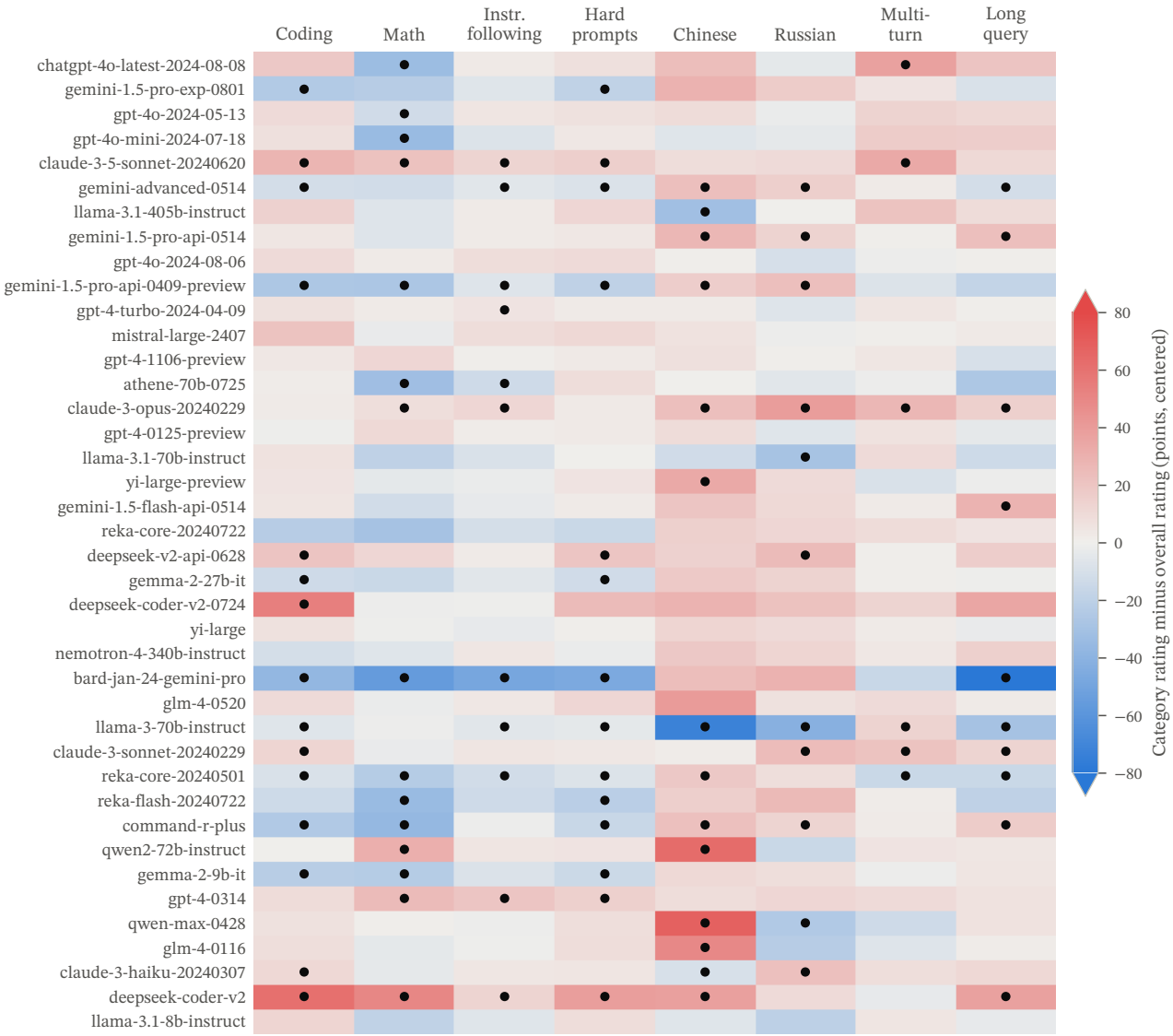


Figure A3: Estimated specialties on the real 2024 board (category rating minus overall rating, centered within the category) of the 40 highest-rated models; dots mark pairs that pass Bonferroni.

F Further limitations and deviations from the proposal

- The truth β^* is itself an estimate and is slightly more spread out than the real strengths, which makes ranking a little easier in the simulation than in reality. The stress tests change the strengths but not the comparison graph or the number of models.
- The normal approximation is weakest for models with few battles; the smallest model in the 2024 data has 1,193 battles.
- In Q3 the overall fit absorbs part of each specialty, most in the largest categories (median null SD 0.81 for instruction following, which contains 35% of battles, against 0.95 for Chinese); an overall fit that leaves the category out would avoid this. An empirical null (Efron, 2004) would be a principled alternative to the bootstrap null.
- BH's FDR guarantee requires assumptions on the null p-values and their dependence that we have not established here. The simulation counts an *attribution error* for a discovery in an unplanted model or a wrong-sign discovery at a planted pair. Other unplanted discoveries in planted models are excluded, without establishing whether their effects are real. BH's mean error share is 0.01–0.12; Bonferroni's probability of any such error is at most 9%. These custom diagnostics do not establish FDR/FWER control: overlap, opponents and refitting can induce effects beyond the planted pairs. Only with no planting is every discovery false: BH FDR is 0.105 (SE 0.022), Bonferroni FWER 0.065 (SE 0.017).
- Not carried out from the proposal: bootstrap rank intervals as a marginal benchmark; BH on the pairwise tests; fully synthetic worlds with other comparison graphs and numbers of models; and a low-rank-plus-sparse decomposition that removes a general-ability component before testing for specialties. We also replaced the proposal's 2025 data sample by the 2024 dump, for the answer key described in Section 2.